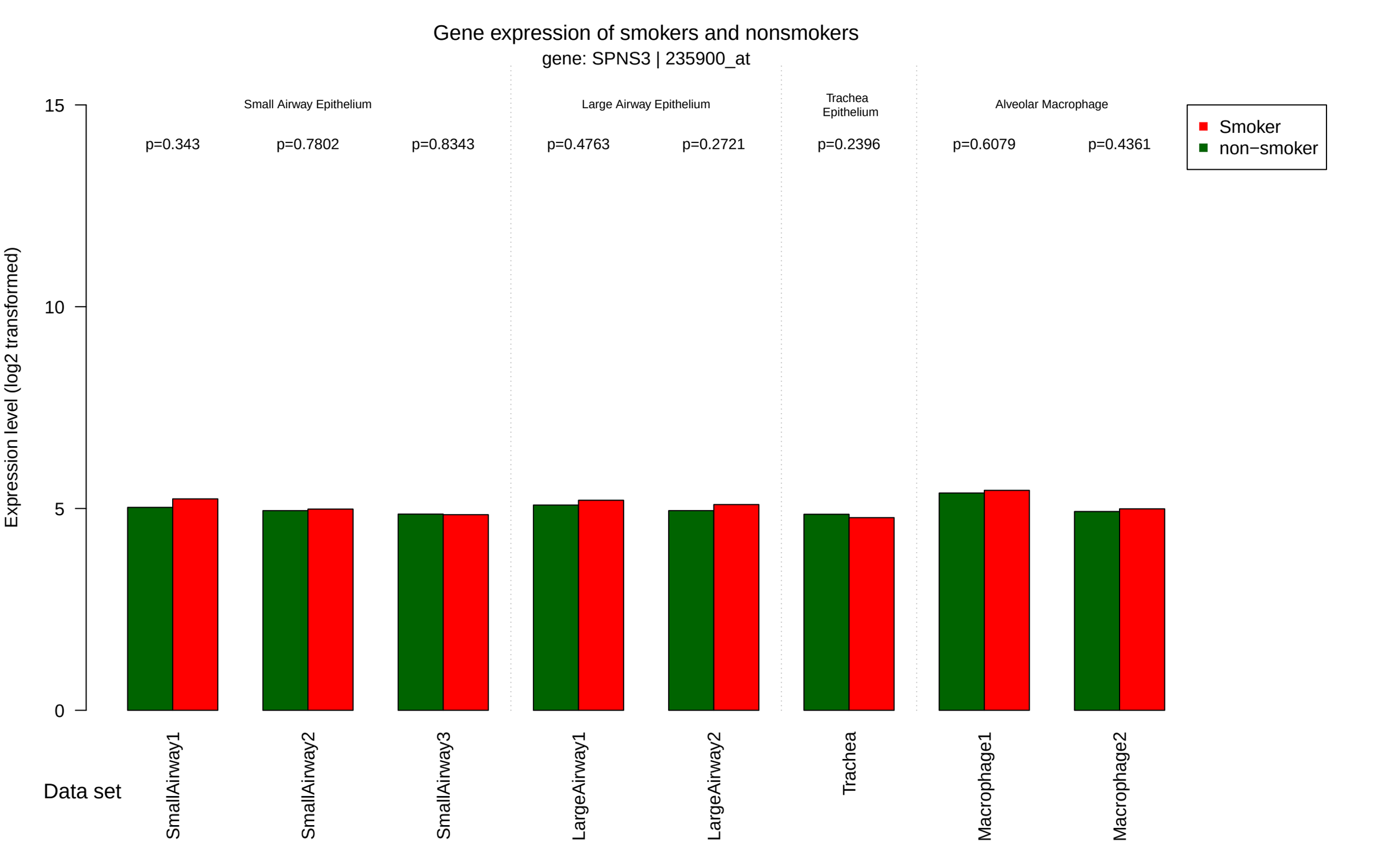
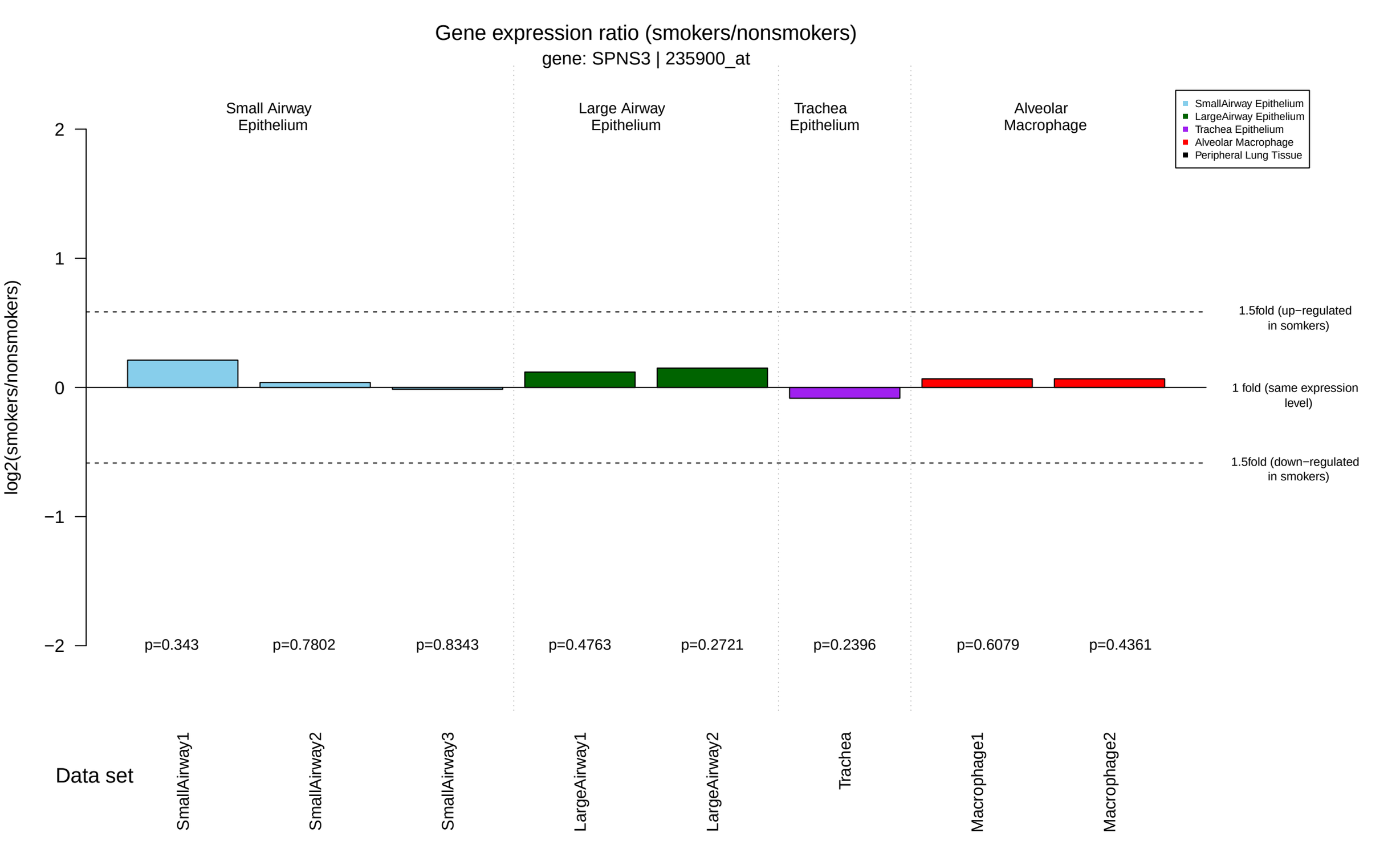
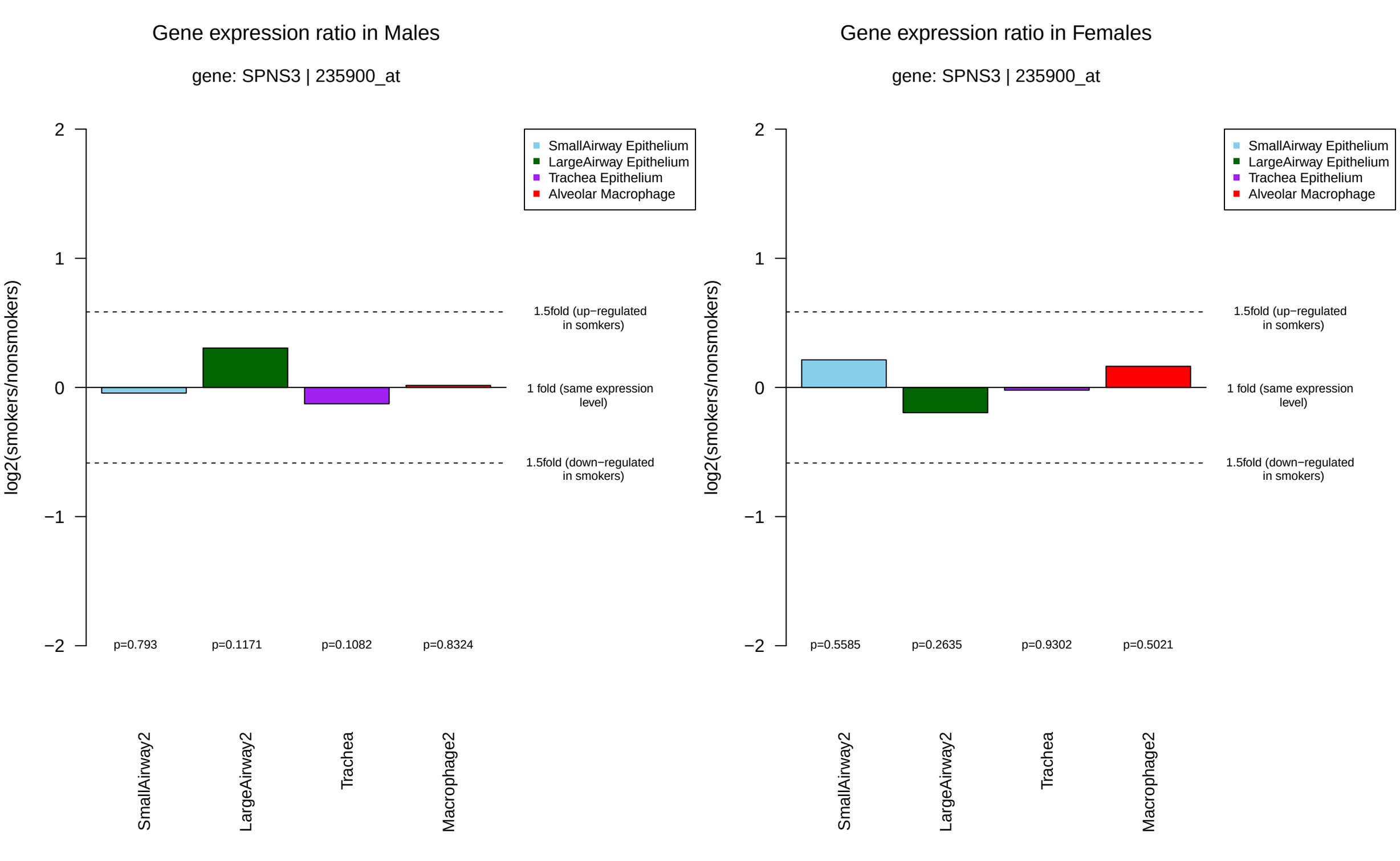
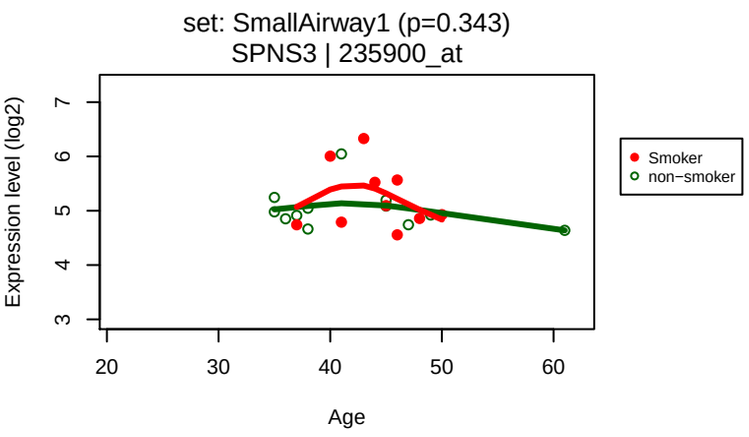
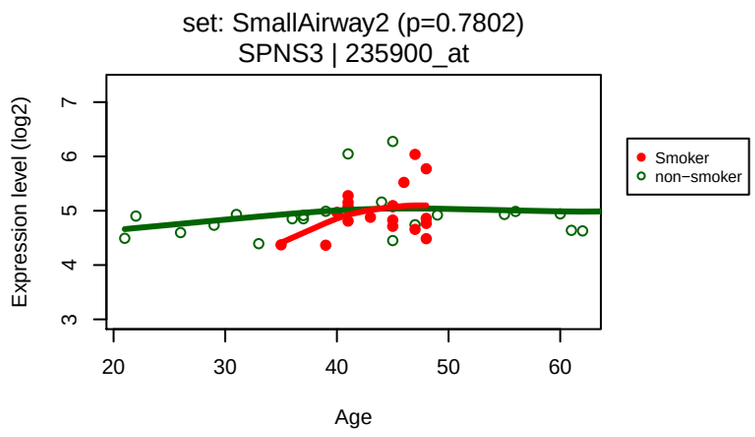
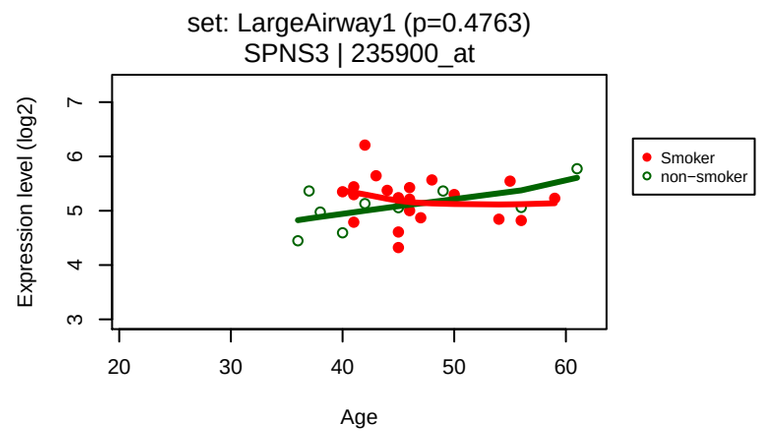

Gene expression of smokers and nonsmokers
gene: SPNS3 | 235900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.343
p=0.7802
p=0.8343
p=0.4763
p=0.2721
p=0.2396
p=0.6079
p=0.4361
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

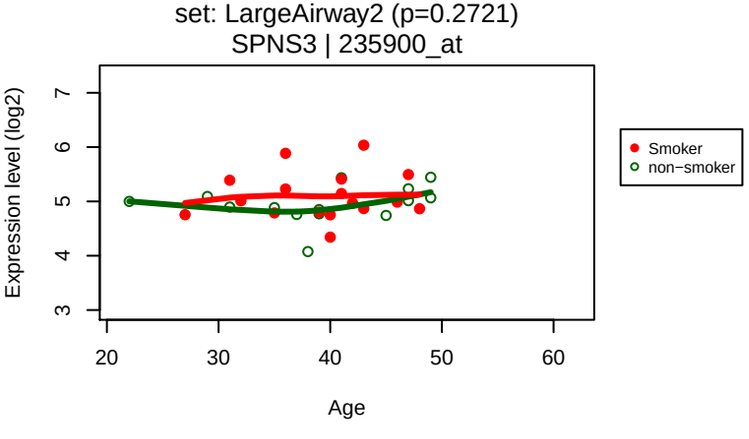

Gene expression ratio (smokers/nonsmokers)
gene: SPNS3 | 235900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.343
p=0.7802
p=0.8343
p=0.4763
p=0.2721
p=0.2396
p=0.6079
p=0.4361
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

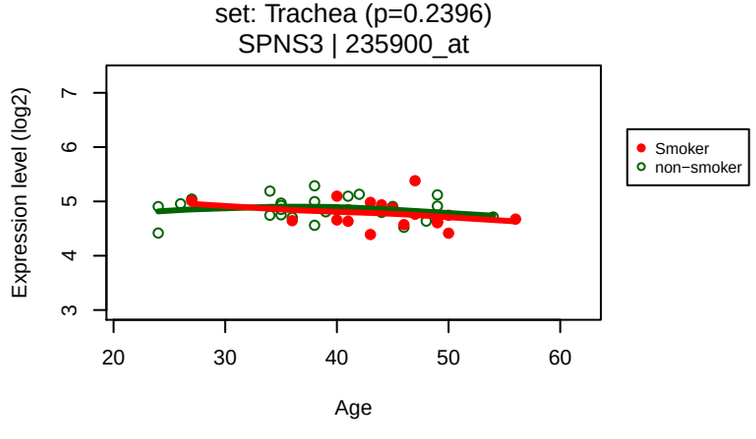

Gene expression ratio in Males
gene: SPNS3 | 235900_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.793
p=0.1171
p=0.1082
p=0.8324
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPNS3 | 235900_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5585
p=0.2635
p=0.9302
p=0.5021
SmallAirway2
LargeAirway2
Trachea
Macrophage2

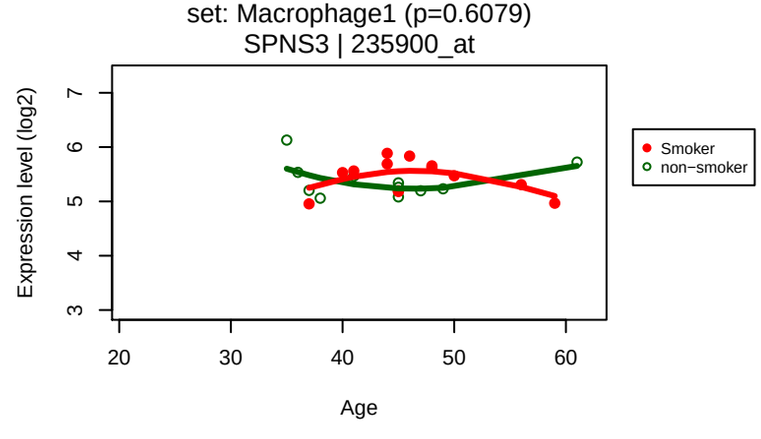

set: SmallAirway1 (p=0.343)
SPNS3 | 235900_at
Expression level (log2)
Age
Smoker
non−smoker

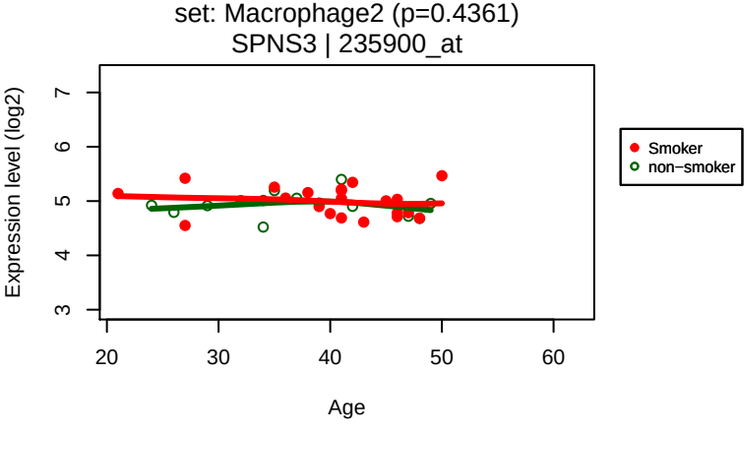

set: SmallAirway2 (p=0.7802)
SPNS3 | 235900_at
Expression level (log2)
Age
Smoker
non−smoker

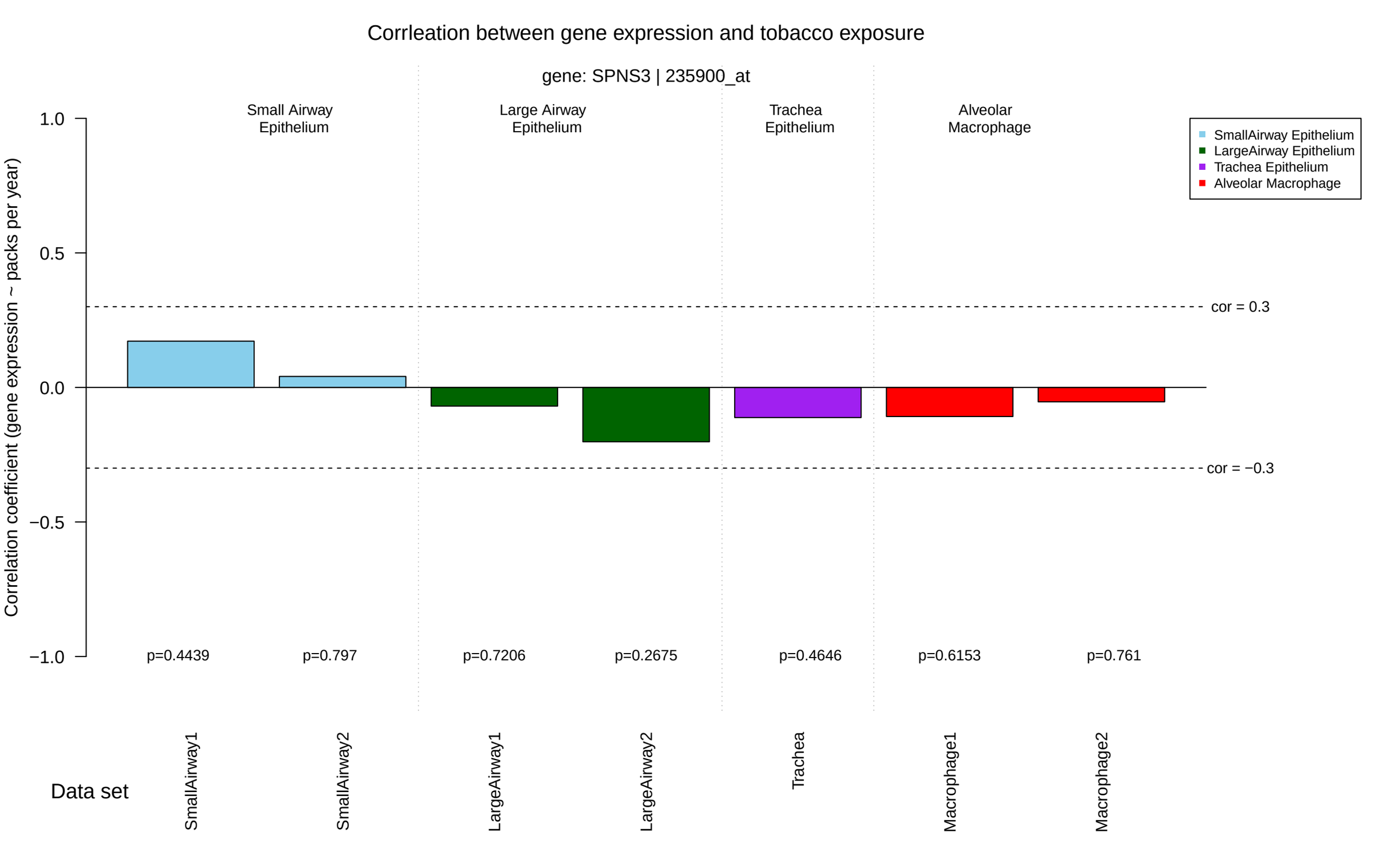

set: LargeAirway1 (p=0.4763)
SPNS3 | 235900_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2721)
SPNS3 | 235900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2396)
SPNS3 | 235900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6079)
SPNS3 | 235900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4361)
SPNS3 | 235900_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPNS3 | 235900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4439
p=0.797
p=0.7206
p=0.2675
p=0.4646
p=0.6153
p=0.761
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2